

Training Program for Software Engineering Major

I. Educational Objectives

This program aims to cultivate high-quality software engineering professionals with well-rounded moral, intellectual, physical, and aesthetic development. Graduates will master fundamental knowledge in software development, demonstrate innovation and teamwork, possess software engineering practice abilities and experience in project development and management, and have strong foreign-language application skills. They will be able to engage in computer software development and software project management in IT enterprises, public institutions, and related organizations, and will be capable of system analysis, development, and project management in application-oriented fields while adapting to technological progress and changing social needs.

II. Training Specifications

Students are expected to love their country, support the leadership of the Communist Party of China, possess a sense of responsibility and social responsibility, have legal awareness and consciously abide by laws and regulations, acquire basic knowledge of humanities, social sciences, and natural sciences, develop learning ability and lifelong learning awareness, maintain good physical and psychological qualities, love the major, and value professional ethics.

The knowledge and abilities are reflected in the following aspects:

1. Master the basic theoretical knowledge and application technologies of computer software and hardware, the fundamental knowledge and theories of software engineering, and advanced software engineering methods, technologies, and tools.
2. Possess strong abilities in software analysis, design, development, and maintenance, as well as preliminary abilities in engineering project organization and management and teamwork.
3. Understand developments in this discipline and related fields, and possess the ability to participate in the organization of computer engineering projects and in software management and development.
4. Possess integrity awareness, team spirit, interpersonal communication skills, and modern awareness.
5. Receive basic training in theoretical thinking and scientific experimentation in the discipline, develop sound ideological and moral qualities, and acquire knowledge through self-study, information retrieval, and communication.
6. Master one foreign language and possess good abilities in foreign-language application, literature retrieval, and scientific and technical writing.

III. Duration of Study: Standard duration is four years.

IV. Degree Awarded: Bachelor of Engineering.

V. Minimum Credits for Graduation: 167 credits.

VI. Principal Discipline: Software Engineering.

VII. Major Courses: JAVA Programming, Discrete Mathematics, Introduction to Software Engineering, Algorithms and Data Structures, Computer Network, Principles of Computer Composition and Assembly Language, Principles of Computer Composition and Assembly Language Practice, Database Principles and Application, Principles of Compiling, Operation System, Engineering Economics, Software Code Development and Human-Computer Interaction Technology, Software Quality Assurance and Testing, Probability Theory and Mathematical Statistics, Object Oriented Methodology, Software Process and Management, Software System Design and System Structure

VIII. Main Practical Teaching Components

1. Comprehensive course design: 12 credits.
2. Major internship and professional practice: 18 credits.
3. Graduation Design: 6 credits.
4. Major practice, competitions, lectures, etc.: 4 credits.

5. Course practice, etc.: 12.67 credits.

IX. Main Major Experiments

1. Basic experiment series: mainly including Introduction to Computer, Advanced Programming Language, etc.

2. Software technology experiment series: mainly including Introduction to Software Engineering, Data Structure experiments, Database Principles and Application experiments, Software Quality Assurance and Testing experiments, Software Code Development and Human-Computer Interaction experiments, Software System Design and System Structure, etc.

3. Hardware technology experiment series: mainly including Computer Network experiments, Operation System experiments, etc.

X. Curriculum

(1) Schedule of General Required Courses

[illegible]

(2) Schedule of Disciplinary Foundation Courses

Course Code	Course Title	Assessment	Credits	Hours	Hour Distribution	Hour Distribution	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester
Course Code	Course Title	Assessment	Credits	Hours	Lecture	Lab/Practice	First Academic Year	First Academic Year	Second Academic Year	Second Academic Year	Third Academic Year	Third Academic Year	Fourth Academic Year	Fourth Academic Year	Semester
Course Code	Course Title	Assessment	Credits	Hours	Lecture	Lab/Practice	1	2	3	4	5	6	7	8	Semester
08319040500	Advanced Mathematics B1	Exam	5	90	90		5								1
08298040250	Advanced Programming Language 1	Exam	2.5	54	36	18	2+1								1
08178050350	Introduction to Computer	Exam	3.5	72	54	18	3+1								1
08239040300	Linear Algebra	Exam	3	54	54		3								1
08320040500	Advanced Mathematics B2	Exam	5	90	90			5							2
08299040250	Advanced Programming Language 2	Exam	2.5	54	36	18		2+1							2
Subtotal	Subtotal	Subtotal	21.5	414	360	54	15	12							

(3) Schedule of Major Courses

Course Code	Course Title	Assessment	Credits	Hours	Hour Distribution	Hour Distribution	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester and Weekly Hours	Semester
Course Code	Course Title	Assessment	Credits	Hours	Lecture	Lab/Practice	First Academic Year	First Academic Year	Second Academic Year	Second Academic Year	Third Academic Year	Third Academic Year	Fourth Academic Year	Fourth Academic Year	Semester
Course Code	Course Title	Assessment	Credits	Hours	Lecture	Lab/Practice	1	2	3	4	5	6	7	8	Semester
08321040350	JAVA Programming	Exam	3.5	72	54	18			3+1						3
08153040400	Discrete Mathematics	Exam	4	72	72				4						3
08322040350	Introduction to Software Engineering	Exam	3.5	72	54	18			3+1						3
08208040450	Algorithms and Data Structures	Exam	4.5	90	72	18			4+1						3
08138040350	Computer Network	Exam	3.5	72	54	18				3+1					4
08310040400	Principles of Computer Composition and Assembly Language	Exam	4	72	72					4					4
08311040100	Principles of Computer Composition and Assembly Language Practice	Assessment	1	36		36				2					4
08195040350	Database Principles and Application	Exam	3.5	72	54	18				3+1					4
08066040350	Principles of Compiling	Exam	3.5	72	54	18					3+1				5
08070040350	Operation System	Exam	3.5	72	54	18					3+1				5
08313040100	Engineering Economics	Exam	1	18	18						1				5
08172040300	Software Code Development and Human-Computer Interaction Technology	Exam	3	72	36	36					2+2				5
08186040250	Software Quality Assurance and Testing	Exam	2.5	54	36	18					2+1				5
08094040300	Probability Theory and Mathematical Statistics	Exam	3	54	54							3			6
08323040300	Object Oriented Methodology	Exam	3	72	36	36						2+2			6
08179040200	Software Process and Management	Exam	2	36	36							2			6
08184040250	Software System Design and System Structure	Exam	2.5	54	36	18						2+1			6
Subtotal	Subtotal	Subtotal	51.5	1062	792	270			17	14	16	12			

(4) Schedule of Major Elective Courses

[illegible]

(5) Schedule of Practical Teaching Components

[illegible]

(6) Distribution of Hours and Credits by Course Category

Category / Item	General Required Courses	General Education Courses	Major Required Courses	Major Required Courses	Major Elective Courses	Practical Teaching Components	Total
Category / Item	General Required Courses	General Education Courses	Disciplinary Foundation Courses	Major Courses	Major Elective Courses	Practical Teaching Components	Total
Hours	720	144	414	1062	216	64 weeks(+552)	2556
Percentage	28.17%	5.63%	16.20%	41.55%	8.45%	/	100%
Credits	34	8	21.5	51.5	12	40(+12.67)	167
Percentage	20.36%	4.79%	12.87%	30.84%	7.19%	31.54%	/